

Describing Welfare

Session 2

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Introduction

Deaton is in your reading/ folder on the hub, or PDF.

- Deaton, chs. 3 and 4
- Deaton & Zaidi, *Guidelines for Constructing Consumption Aggregates*
 - Deaton & Zaidi §§2–4 if you read nothing else
- **Christiaensen, Ligon, & Sohnesen (2022)**: at least memorize the title.

Measuring Individual Welfare

Many notions of individual welfare; generally lead to some unidimensional index, say w (bigger is better):

- Subjective ("How happy are you?")
- Utility (How is this different from Subjective happiness?)
- Material (Something about command over economic resources)
 - Wealth?
 - Income?
 - Consumption expenditures?

Aside: What about risk?

Material Well-Being: Statistics

Some summary statistic of the *distribution* F of w in the population.

Inequality E.g., Atkinson's (1971) utilitarian notion of inequality:

$$I = U \left(\int_{\underline{w}}^{\bar{w}} w dF(w) \right) - \int_{\underline{w}}^{\bar{w}} U(w) dF(w) \geq 0$$

Poverty A quantile of some function $\phi_z(w)$; e.g.,

$$P(z) = \int_{\underline{w}}^z \phi_z(w) dF(w)$$

These are distributions *of the population*, so in practice those sample weights will matter!

Conventional options for w

Income

- Difficult to measure if sources are agricultural or informal
- Also highly variable—a farmer's main income may arrive once a year, after harvest, and be very risky.

Wealth

- What is it?
- Perhaps something like discounted expected income?
- But if income is difficult to measure then all possible future incomes may be much more difficult

Conventional options for w , continued

Assets

- Related to wealth, but may be measurable now
- Financial assets + Bicycles + Goats + House = ?? In principle these could be priced, but this may not be at all practical.

Consumption Expenditures

The "Gold standard"

- Probably easier to measure than income
- Consumption is smoother than income
- And closer to the theoretical object (permanent income, lifetime resources) that we really want.

The Theoretical Object

The household's problem

What is the consumption aggregate a measure of? Assume:

- (A1) Preferences** $U(c)$ over J goods, increasing, strictly concave, the same every period, discounted at β .
- (A2) One decision maker** the household maximizes a single U .
- (A3) One asset** a risk-free bond, gross return R , no borrowing limit.
- (A4) Expectations** $\mathbb{E}_t$ over the true conditional distribution.

With A_t resources on hand and $x_t = p_t \cdot c_t$ total expenditure,

$$V_t(A_t, p_t) = \max_{c_t} U(c_t) + \beta \mathbb{E}_t V_{t+1}(A_{t+1}, p_{t+1}),$$

subject to $A_{t+1} = R(A_t - x_t) + y_{t+1}$.

The marginal utility of expenditure

Differentiate. With $u_j = \partial U / \partial c_j$, the first-order condition in c_j and the envelope condition in A are

$$u_j(c_t) = \lambda_t p_{tj}, \quad \lambda_t \equiv \beta R \mathbb{E}_t[\partial V_{t+1} / \partial A] = \partial V_t / \partial A.$$

So λ_t is two things at once, which is what makes it useful:

- the multiplier on the period budget constraint, the utility value of one more cedi spent *today*;
- the derivative of the *value function*, the utility value of one more cedi of resources, the whole future included.

V is concave in A , so λ falls as the household gets richer.

Write $w_t = -\log \lambda_t$, which therefore rises with welfare. This is another candidate for the w of the opening slide.

The martingale condition

Substituting the envelope condition into the definition of λ_t ,

$$\lambda_t = \beta R \mathbb{E}_t \lambda_{t+1}.$$

When $\beta R = 1$ the marginal utility of expenditure is a martingale:

$$\mathbb{E}_t \lambda_{t+1} = \lambda_t.$$

A statement about information: whatever the household knew at t is already in λ_t , so nothing dated t forecasts the change.

- Needs A1–A4 *and* an interior solution: no binding borrowing limit.
- With several assets, $\beta \mathbb{E}_t[(\lambda_{t+1}/\lambda_t) R_{t+1}^j] = 1$ for every asset j held in positive quantity.

Why total expenditure?

Three steps, each with its own assumption. Let

$v(x, p) = \max\{U(c) : p \cdot c \leq x\}$, so $\lambda = \partial v / \partial x$.

1. At *given* prices x orders welfare, since $\lambda > 0$. Nothing beyond A1 is needed, but this ranks within a market only.
2. To rank across markets we need a scalar $P(p)$ with $v(x, p) = \psi(x/P(p))$. Such a P exists if and only if U is homothetic (A5), and then $P = e(p)$, the cost of a unit of the aggregate. Quasi-homothetic (A5'): two indices, a line $a(p)$ and a deflator $b(p)$.
3. So $\lambda = \psi'(x/e(p))/e(p)$, decreasing in real expenditure.

Under A5, real total expenditure $x/e(p)$ is a sufficient statistic for within-period welfare. It is what the rest of this session builds.

Hall's random walk

The strict permanent income hypothesis adds:

(A8) Constant prices or a martingale in p independent of λ , which puts the walk in quantities.

(A9) Quadratic utility $U(x) = -\frac{1}{2}(\bar{x} - x)^2$, so $\lambda_t = \bar{x} - x_t$ is linear in expenditure.

(A10) $\beta R = 1$ with R constant and known.

The martingale in λ is then a martingale in x :

$$\mathbb{E}_t[\bar{x} - x_{t+1}] = \bar{x} - x_t \implies \mathbb{E}_t x_{t+1} = x_t.$$

Consumption is a random walk (Hall 1978): nothing dated t forecasts Δx_{t+1} , lagged income included. Testable, but it needs longitudinal data.

The assumptions, collected

	Assumption	What it buys
A1–A2	Time-separable U , one maximizer	a well-defined λ
A3	One asset, no borrowing limit	an interior solution
A4	Rational expectations	$\mathbb{E}_t$ is the truth
A5	Homotheticity, or quasi-	a deflator, or a line and one
A6	Iso-elastic cardinalization	w affine in log real x
A7	Durables weakly separable	nondurables and services
A8–A10	Constant p , quadratic U , $\beta R = 1$	Hall's random walk

Everything from here to the end of the session is an attempt to measure x . None of it repairs an assumption in this table.

Practical Difficulties in Measuring Individual Items of Consumption

The four components of consumption expenditures

Or: "Nondurable consumption and services".

Consumption expenditures in the LSMS surveys generally fall into one of five categories, and are typically elicited in different modules.

1. **Food** — purchased, own-produced, and received as gifts
2. **Non-food, non-durable** — fuel, transport, clothing, school fees, health
3. **Durables** — *value of services*, or "use value", not purchase price
4. **Housing** — rent, actual or imputed
5. **Other Services** — Not really measured at all

List Length

Beegle et al (2012): the *number* of goods asked has a large effect on recall, and on that evidence many NSOs have lengthened the list. Just for food:

Ethiopia	25 → 26 → 55 → 73 → 74
GhanaLSS	58 → 58 → 117 → 112 → 114 → 114 → 179
Burkina Faso	55 → 130 → 152
Uganda	61 → 63 → 70 → 71 → 77 → 77 → 117 → 116
Nigeria	90 → 90 → 96 → 104 → 116 → 116 → 110 → 110

- Should make measured total consumption more accurate;
- BUT: Really complicates comparisons across time.
- And: Can greatly increase cost of fielding the survey.

Timing & Recall

The model treats time as discrete. What in the field corresponds to a "period"? LSMS practice reveals what the designers thought.

Food Usually a seven-day recall, with interesting variation and some explicit experiments (Beegle et al. 2012) on how well respondents remember purchases.

Non-food expenditures or "semi-durables" Think things like clothing, inventories of kerosene.... Often a 30-day recall.

Durables Bicycles, radios, fans,... Often a year-long recall for purchases, and/or an inventory of the *stock* of durables.

Food prices: which price?

Food *expenditures* are quantities times prices. Which prices?

- farmgate price received — what the household gave up
- local market price — what it would have cost
- these differ by the marketing margin, said to be 30–50%

Deaton & Zaidi: value at the price the household *faces*. In practice, a median unit value by item, cluster and round, falling back up the hierarchy when cells are thin.

Which of the two does a survey's own valuation question measure?

Food prices: what GLSS7 asks, and what it gets

Section 8 Part H, question 9: **"for how much would you sell one unit of ... now?"** A farmgate question, by its wording.

Against what neighbours *paid*, same item, unit and cluster:

- Ghana 2016-17, 322 cells: median ratio **1.000**; a third tie exactly
- Uganda, eight waves, the same question: **1.000** pooled
- Uganda also records sales: the valuation is **1.23** times the sale price; buyers paid **1.60** times it

Asked what a bowl of maize would sell for, a household answers with the price of a bowl of maize. The margin is real — 1.3 to 2.2 where a survey records sales — and this question does not measure it.

Food quantities, and what was never bought

Much of the food in an LSMS household was never bought:

- Purchased at market
- Consumed out of own production
- Gift/Transfer
- Consumed outside the household (?)

Not a canard. In Ghana 2016-17 own production is **28.7%** of the value of food; 12,191 of 13,942 households record some. Deaton and Zaidi's Table 3.1 implies 31.9%.

But it reaches your aggregate only if you *value* it: from 1991-92 on the survey records own production in quantities, not cedis.

→ notebook

GLSS7's answer: a diary, visited six times

The most recent GhanaLSS is ambitious about frequent purchases: a diary, and an enumerator visiting six times at five-day intervals.

The form (reading/): six visit columns headed *2nd* to *7th*, asking "...**since my last visit?**", and no price column anywhere. Expensive; two experiments have asked whether it is worth it.

Scott & Amenuvegbe (1990) Ghana, and the reason GLSS asks this way

Beegle et al. (2012) Tanzania, eight modules, randomly assigned

The diary

CLUSTER NUMBER:

HOUSEHOLD NUMBER:

NAME OF HOUSEHOLD HEAD: DATE:

A R DAY MTH Y E

INSTRUCTION: RECORD ALL PAYMENTS MADE AND ESTIMATE ALL OWN PRODUCED GOODS CONSUMED.

Date Amount	Place of Purchase/ Own Produce	Description of Item Purchased or	Consumed	Qty	Unit	Price

"Record all payments made and *estimate all own produced goods consumed*": the diary asks for a price on own produce. Section 9B, filled from it, does not.

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How fast does recall decay?

Scott & Amenuvegbe (1990), 13 frequently purchased items in Ghana:

- Reported daily expenditure falls steadily as the recall window lengthens from one day to one week, then flattens between one and two weeks
- Roughly **2.9% lost for every day added**, after reweighting away purchase cycles and start-up bias
- The decline is larger for items bought more often
- A one-year recall is simply erratic

So a weekly recall understates frequent purchases by around **20%** — and that was, in their words, "an abnormally well controlled field situation."

What actually drives the error

Beegle et al. (2012): eight modules randomly assigned across ~4,000 Tanzanian households, benchmarked against a personal diary.

- Recall *length* is the small knob: 7 vs 14 days moves food spending **7%**
- Item *list* design is the big one: **21%** between 58 items and 11 categories
- "Usual month" fails outright, and contaminates the module after it
- Household diaries miss about **19%** of what personal diaries catch

Same households: poverty from **47.5% to 66.8%**, by module alone.

Timing: what is a period?

For food, a day, probably: people schedule around meals. Surveys pick something longer, and each choice buys one error with another.

- A *diary* of what was eaten, meal by meal, is the ideal and the burden: hard to sustain, and paying for it buys a record, not an accurate one
- A *short* window is cheap but noisy: day-to-day variation in intake is not the error we are after
- *Usual* consumption asks the respondent to do our averaging for us, and to share our idea of "typical"; Beegle et al. found it fails outright

Inventories: purchases are not consumption

Storable staples — dry maize, rice, oil — are bought in bulk and drawn down. Perishables — fruit, fresh fish — are bought as eaten.

- A short window catches the fruit and misses the maize sack, or catches the sack and triples the week
- Averages out across households if purchase timing is unrelated to the visit; does not average out *within* one, which is where the line is applied
- Who buys in bulk is a matter of wealth: cash on hand, a dry store, a refrigerator. So the noise is on the rich, and on the staples
- Thirty days of visits (GLSS7) is one answer; asking what was *consumed*, as 8H does and 9B does not, is the other

Price Indices

Two different price indices, meant to address two different problems:

- **Temporal:** prices rise between rounds. Use the CPI, or better, a survey-based Paasche index built from the survey's own unit values.
- **Spatial:** prices differ across regions *within* a round, often by more than they differ across a decade. A national poverty line applied to undeflated nominal consumption will find poverty wherever prices are high.

A household-specific Paasche index

Deaton & Zaidi recommend a household-specific Paasche index

$$P_i = \left[\sum_j s_{ij} \left(\frac{p_{ij}}{p_j^0} \right)^{-1} \right]^{-1},$$

with s_{ij} household i 's budget share on item j .

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Durables and housing

Durables: what enters consumption is the *flow of services*, not the purchase. For a stock S with service life L and interest rate r ,

$$\text{use value} \approx S \left(r + \frac{1}{L} \right).$$

Housing: for owner-occupiers there is no rent, so impute it.

- hedonic regression of log rent on housing characteristics, among renters
- predict for owners
- fragile where the rental market is thin, which is most rural areas

Intra-Household Allocation

Adult equivalence

A household of six is not six times as well off as a household of one with the same total consumption. Two adjustments, usually conflated:

- **Household composition:** children cost less than adults —
 $\Rightarrow$ adult-equivalent scale $A_i = \sum_k a_{\text{age},\text{sex}}$
- **Economies of scale:** shared goods, so cost rises less than proportionately — $\Rightarrow n_i^\theta$, with $\theta \in (0, 1)$

Combined: $c_i = \frac{C_i}{A_i^\theta}$.

θ is not estimated in this session; it is *assumed*. Report the poverty rate for several values of it.

Barten: children re-price goods

A^θ makes a child a fraction of an adult *for everything*. Barten (1964): composition changes *which* goods a household needs,

$$\max_c U\left(\frac{c_1}{m_1(z)}, \dots, \frac{c_J}{m_J(z)}\right) \quad \text{s.t.} \quad \sum_j p_j c_j = x,$$

so a household with demographics z is the reference household at prices $p_j m_j(z)$. A child makes milk dear; substitution does the rest.

- the scale is a ratio of cost functions, $e(u, p \circ m(z))/e(u, p)$: it depends on prices, and in general on u
- A^θ is the case where every m_j is the same number: children change the *size* of the bundle, never its *shape*
- a shape that depends on z is not what A5 allows

Constructing Aggregate Welfare Measures

Poverty measures

The Foster–Greer–Thorbecke class: for poverty line z ,

$$P_{\alpha} = \frac{1}{N} \sum_{i:c_i < z} \left(\frac{z - c_i}{z} \right)^{\alpha}.$$

- $\alpha = 0$: headcount ratio. Ignores *how* poor the poor are.
- $\alpha = 1$: poverty gap. The transfer needed, per capita, as a share of z .
- $\alpha = 2$: squared gap. Weights the poorest most; the only member of the family that satisfies the transfer axiom.

Weighted, always: $P_{\alpha} = \sum_i w_i \left(\frac{z - c_i}{z} \right)^{\alpha} / \sum_i w_i$.

→ notebook

- **Lorenz curve:** cumulative share of consumption against cumulative share of population. Everything else is a summary of it.
- **Gini:** twice the area between the Lorenz curve and the 45° line. Bounded, familiar, and insensitive where it matters most.
- **Generalized entropy** $GE(\alpha)$: decomposable into within- and between-group parts. $GE(0)$ is mean log deviation, $GE(1)$ is Theil.

Inequality: Atkinson

The opening slide's $I = U(\int w dF) - \int U(w) dF$ is in utils, which nobody reports. Let w_{ede} solve $U(w_{\text{ede}}) = \int U dF$: the level which, given to everyone, yields the same total utility as the actual spread. Then $I = U(\bar{w}) - U(w_{\text{ede}})$, and

$$A = 1 - \frac{w_{\text{ede}}}{\bar{w}}$$

is the same judgement in cedis per cedi, with $U(w) = w^{1-\varepsilon}/(1-\varepsilon)$ and ε assumed, like θ .

$A = 0.3$: this society would give up 30% of mean consumption to have the rest shared equally. Prefer the Lorenz curve to any scalar.

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One index, or several?

Everything here divides nominal expenditure by a scalar: a Paasche index over space, a CPI over time, an equivalence scale over composition. Three scalars, one household, one number out.

- one deflator is exact at **rank one**: budget shares do not vary with total expenditure
- but they do — Engel's law, which opened this session
- a line *and* a deflator are exact at **rank two** (Gorman), which Engel's law permits
- so on what grounds is a two-index procedure adequate?

Hold the question: session 3 answers it for Uganda.

Exercises

1. The poverty line used above is fake. Build a real one: take the food bundle consumed by households in the second and third deciles, price it at national median unit values, and scale it to 2,900 kcal per adult equivalent.
Recompute P_0, P_1, P_2 . Notebook: `poverty_line.ipynb`
2. Deflate spatially. Construct a regional Paasche index from the survey's own unit values, apply it, and report how much of the north-south poverty gradient survives.
Notebook: `spatial_deflation.ipynb`
3. Vary θ in $c_i = C_i/A_i^\theta$ over $[0.5, 1.0]$ and plot the headcount ratio against it. Over what range does the *ranking* of regions change? Notebook: `equivalence_scales.ipynb`